

UNIT 4

MATTER AND ITS PROPERTIES

Learning objectives:

- Understand the properties of solids, liquids, and gases based on particle arrangement, shape, and volume.
- Explain how the particle theory relates to changes in states of matter.
- Describe state changes (melting, freezing, boiling, condensation, sublimation) and the role of heat.



MATTER AND ITS PROPERTIES

Introduction to Matter

Everything around us, including the air we breathe, the water we drink, and the objects we use, is made of **matter**. Matter is anything that has **mass** and takes up **space**.

States of Matter

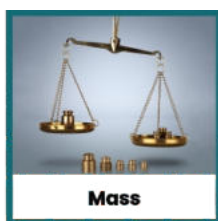
Matter exists in three main states:

1. **Solid** – It has a fixed shape and volume (e.g., wood, ice, stone).
2. **Liquid** – It takes the shape of the container but has a fixed volume (e.g., water, milk, oil).
3. **Gas** – It has no fixed shape or volume (e.g., air, steam, oxygen).

Properties of Matter

Matter has different properties that help us identify and classify it.

1. **Mass**: The amount of matter in an object (measured in grams or kilograms).
2. **Volume**: The space an object occupies (measured in litres or cubic metres).
3. **Shape**: Some objects have a fixed shape (solids), while others take the shape of their container (liquids and gases).
4. **Texture**: How an object feels (rough, smooth, soft, or hard).
5. **Solubility**: Some substances dissolve in water (like salt and sugar), while others do not (like sand and oil).
6. **Transparency**: Some materials allow light to pass through (transparent), some allow partial light (translucent), and some block light completely (opaque).



Mass



Volume



Shapes



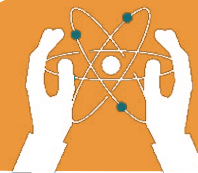
Texture



Solubility



Transparency



Particle: A tiny unit of matter such as atoms or molecules.

Properties of Solids, Liquids, and Gases

a. Solids

- Particles in solids are tightly packed in a regular pattern.
- Solids have a definite shape and volume.
- Particles can only vibrate in place.

b. Liquids

- Particles in liquids are closely packed but not in a fixed pattern.
- Liquids have a definite volume but take the shape of their container.
- Particles can move around each other.

c. Gases

- Particles in gases are far apart and randomly arranged.
- Gases have no definite shape or volume.
- Particles move freely in all directions.

Particle Theory of Matter

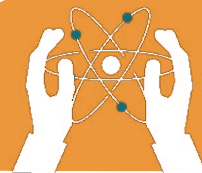
Scientists have developed a model to explain the behaviour of matter:

- All matter is made of tiny particles.
- There are spaces between the particles of matter.
- Particles attract each other.
- Particles are always in motion.
- Adding heat makes particles move faster.

● DID YOU KNOW?

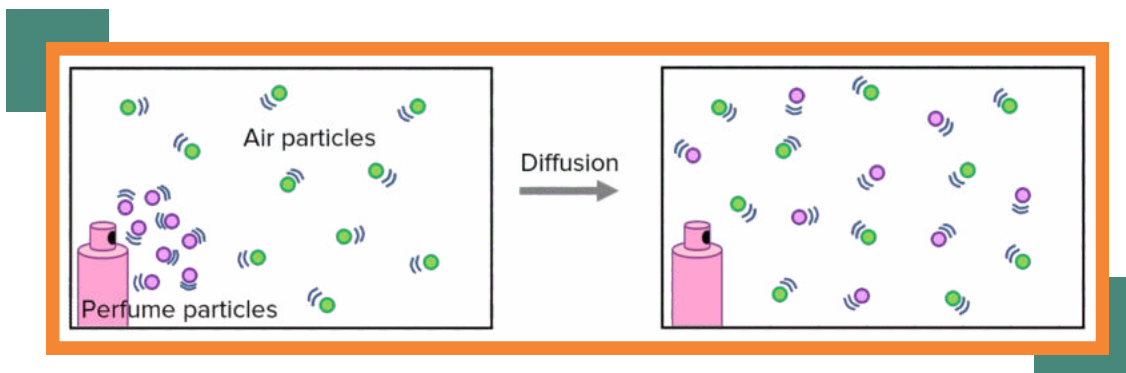


The sun and stars are actually in a fourth state of matter called plasma, where gas particles are so energized that they lose electrons and create a charged state.



Diffusion of Particles

Diffusion is the movement of particles from an area of high concentration to an area of low concentration. Gases diffuse the fastest as compared to liquids.



Experiment

Question:

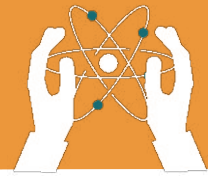
Spray perfume or air freshener in one corner of the room. After a few seconds, can you smell it even if you are far away? How do the particles spread through the air?

Changes in state of Matter:

Changes in matter occur due to the behaviour and interactions of tiny particles (atoms and molecules) that make up substance.

State conversion of matter

1. Melting: It is the process where a solid changes into a liquid by adding heat. When heat is applied, the particles in the solid begin to move faster, weakening the forces holding them together, which allows the solid to become a liquid.



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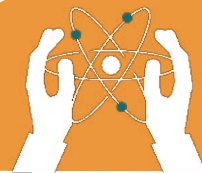


2.Freezing: It is the transformation from a liquid to a solid by removing heat. As heat is removed, the particles in the liquid slow down, allowing them to come closer together and form a solid structure.

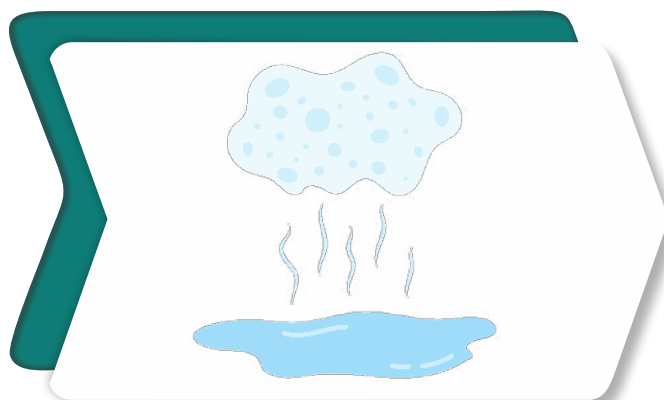


3.Boiling: It is the process where a liquid changes into a gas by adding heat. When heated, the particles in the liquid move rapidly, eventually escaping into the air as gas.





4. Condensation: It is the process where a gas changes into a liquid by removing heat. When gas particles lose heat, they slow down and come closer together, forming a liquid.



5. Sublimation: It is the process where a solid changes directly into a gas without becoming a liquid, typically by adding heat. Certain substances, like dry ice (solid CO_2), go directly from solid to gas when heated.



Melting Ice Experiment

Place ice cubes in a bowl.

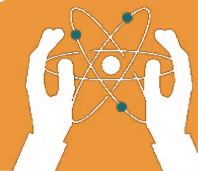
Observe the process of melting and note the temperature at which it happens.



MATTER AND ITS PROPERTIES

State Change	Initial State	Final State	Heat Added/ Removed
Melting			
Freezing			
Boiling			
Condensation			
Sublimation			

MATTER AND ITS PROPERTIES



Match the Column.

COLUMN A

Melting

Freezing

Boiling

Condensation

COLUMN B

Liquid to Gas

Gas to Liquid

Solid to Liquid

Liquid to Solid

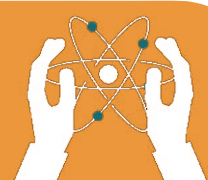
Project Idea

Scientific Investigation on States of Matter:

Show different types of changes in matter (reversible and irreversible changes) with simple experiments.

Questions:

- What happens when ice melts? Is this change reversible?
- What happens when paper is burned? Is this change reversible or irreversible?
- How does cooking change the state of food?



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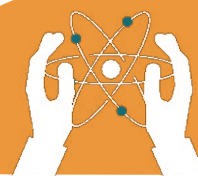
Answer the Following Questions:

Short-Answer Questions

- Q1. Define matter and also name the states of matter.
- Q2. Explain the particle theory of matter.
- Q3. Why do gases not have a definite shape or volume?

Long-Answer Questions

- Q1. Explain the properties of solids, liquids and gases.
- Q2. How do the properties of solids, liquids, and gases influence fire safety methods, such as fire blankets, Co_2 cylinders, and foam extinguishers?



Reasoning Questions

1. Why can gases be compressed more easily than solids or liquids?

Ans: _____

2. Why does a liquid take the shape of its container but retain its volume?

Ans: _____

3. Why does heating a solid eventually turn it into a liquid?

Ans: _____



Critical Thinking

Consider why substances like dry ice skip the liquid state and go directly from solid to gas. How might this be useful in real-life applications?